LOAN APPROVAL PREDICTION

InternSpark Data Science Internship

Student Details

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Internship Role: Data Science Intern

Project Title: Loan Approval Prediction

Tools Used: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Google Colab

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Introduction

Loan approval is an important process in the banking and financial sector. Financial institutions receive numerous loan applications every day and must evaluate applicants carefully before approving loans. Manual evaluation can be time-consuming and may lead to inconsistencies. Machine learning can help automate this process by identifying patterns in historical loan data and predicting whether a loan application is likely to be approved.

The objective of this project was to build a machine learning model that predicts loan approval based on applicant information such as income, education, employment status, loan amount, credit history, and property area. The project also focused on handling missing values, preprocessing data, addressing class imbalance, and evaluating model performance using different metrics.

Dataset Overview

The dataset contains information about loan applicants and their loan approval status.

Dataset Details

- Number of Records: 614
- Number of Columns: 13

Key Attributes

- Loan ID
- Gender
- Married
- Dependents
- Education
- Self Employed
- Applicant Income
- Coapplicant Income
- Loan Amount
- Loan Amount Term
- Credit History
- Property Area
- Loan Status

The target variable in this dataset is **Loan Status**, which indicates whether a loan application was approved or rejected.

Data Cleaning and Preparation

Before building the machine learning models, the dataset was cleaned and prepared.

The following preprocessing steps were performed:

- Inspected the dataset structure and data types.
- Identified missing values in multiple columns.
- Filled categorical missing values using the mode.
- Filled numerical missing values using the median.
- Encoded categorical variables using Label Encoding.
- Scaled numerical features using StandardScaler.
- Checked class distribution in the target variable.
- Applied SMOTE (Synthetic Minority Oversampling Technique) to balance the classes.

After preprocessing, the dataset contained no missing values and was ready for model training.

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property_Area	Loan_Status
0	LP001002	Male	No	0	Graduate	No	5849	0.0	NaN	360.0	1.0	Urban	Y
1	LP001003	Male	Yes	1	Graduate	No	4583	1508.0	128.0	360.0	1.0	Rural	N
2	LP001005	Male	Yes	0	Graduate	Yes	3000	0.0	66.0	360.0	1.0	Urban	Y
3	LP001006	Male	Yes	0	Not Graduate	No	2583	2358.0	120.0	360.0	1.0	Urban	Y
4	LP001008	Male	No	0	Graduate	No	6000	0.0	141.0	360.0	1.0	Urban	Y

```
<class 'pandas.core.frame.DataFrame'>
Index: 614 entries, 0 to 613
Data columns (total 13 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Loan_ID            614 non-null    object
 1   Gender             601 non-null    object
 2   Married            611 non-null    object
 3   Dependents         599 non-null    object
 4   Education          614 non-null    object
 5   Self_Employed      582 non-null    object
 6   ApplicantIncome    614 non-null    int64
 7   CoapplicantIncome  614 non-null    float64
 8   LoanAmount         592 non-null    float64
 9   Loan_Amount_Term   600 non-null    float64
 10  Credit_History     564 non-null    float64
 11  Property_Area      614 non-null    object
 12  Loan_Status        614 non-null    object
dtypes: float64(4), int64(1), object(8)
memory usage: 67.2+ KB
```

Class Distribution Analysis

The distribution of loan approvals was analyzed before applying SMOTE.

Loan Status	Count
Approved	422
Rejected	192

The dataset showed moderate class imbalance, with approved loans occurring more frequently than rejected loans.

After applying SMOTE, both classes were balanced.

Loan Status	Count
Approved	342

Rejected	342

```
Loan_Status
Y    422
N    192
Name: count, dtype: int64
```

```
Loan_Status
0    342
1    342
Name: count, dtype: int64
```

Model Development

Two machine learning models were trained and evaluated:

1. Logistic Regression

Logistic Regression was used as a baseline classification model due to its simplicity and interpretability.

2. Random Forest Classifier

Random Forest was used as a tree-based ensemble model capable of capturing more complex relationships within the data.

The dataset was split into training and testing sets before model training.

```
(491, 12)
(123, 12)
```

Model Evaluation

The models were evaluated using the following metrics:

- Precision
- Recall
- F1 Score
- ROC-AUC Score

Logistic Regression Results

Metric	Value
Precision	0.768
Recall	0.913
F1 Score	0.834
ROC-AUC	0.700

```
Accuracy : 0.7886178861788617
Precision : 0.7596153846153846
Recall : 0.9875
F1 Score : 0.8586956521739131
ROC AUC : 0.7030523255813954
```

Random Forest Results

Metric	Value
Precision	0.758
Recall	0.900
F1 Score	0.823
ROC-AUC	0.683

```
Accuracy : 0.7804878048780488
Precision : 0.7572815533980582
Recall : 0.975
F1 Score : 0.8524590163934426
ROC AUC : 0.6968023255813953
```

Model Comparison

Model	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.768	0.913	0.834	0.700
Random Forest	0.758	0.900	0.823	0.683

The results indicate that Logistic Regression performed slightly better than Random Forest across all evaluation metrics.

```
                 Model  Precision  Recall  F1 Score   ROC AUC
0  Logistic Regression   0.759615  0.9875  0.858696  0.703052
1        Random Forest   0.757282  0.9750  0.852459  0.696802
```

Key Findings

1. The dataset contained 614 loan applications.
2. Missing values were successfully handled during preprocessing.
3. The dataset initially showed class imbalance between approved and rejected loans.
4. SMOTE was used to balance the target classes.
5. Logistic Regression achieved the best overall performance.
6. Logistic Regression obtained an F1 Score of 0.834.
7. Logistic Regression achieved a ROC-AUC score of 0.700.
8. Credit history and applicant-related financial information are important indicators of loan approval.
9. Balanced data improved fairness in model training.
10. Machine learning can support faster and more consistent loan approval decisions.

Business Interpretation

The developed model can help financial institutions identify applicants who are more likely to receive loan approval. By automating the initial screening process, banks can reduce processing time and improve operational efficiency.

The model should be used as a decision-support tool rather than a complete replacement for human review. Loan officers can use the model’s predictions alongside traditional evaluation procedures to make informed decisions.

Deployment Recommendation

Based on the evaluation results, Logistic Regression is recommended for deployment because it provides the highest overall performance while remaining simple, interpretable, and easy to maintain.

A default classification threshold of 0.50 can be used initially. However, financial institutions may adjust the threshold depending on their risk tolerance and business objectives.

Executive Summary

This project focused on building a machine learning model to predict loan approval outcomes using borrower information. The dataset included demographic, financial, and credit-related information about loan applicants. Data preprocessing techniques were applied to handle missing values, encode categorical variables, and scale numerical features.

To address class imbalance, SMOTE was used to generate a balanced training dataset. Two machine learning models, Logistic Regression and Random Forest, were trained and evaluated using Precision, Recall, F1 Score, and ROC-AUC metrics.

The results showed that Logistic Regression achieved the best overall performance, making it the preferred model for deployment. The project demonstrates how machine learning can assist financial institutions in making faster, more consistent, and data-driven loan approval decisions.

Conclusion

The Loan Approval Prediction project successfully demonstrated the use of machine learning techniques for solving a real-world financial problem. Through preprocessing, class balancing, model training, and evaluation, meaningful insights were obtained from the dataset.

Among the evaluated models, Logistic Regression delivered the strongest performance and is recommended for deployment. The project highlights the value of predictive analytics in improving decision-making processes within the banking and finance sector.

Future improvements may include testing additional machine learning algorithms, performing feature engineering, and deploying the model as a web-based application for real-time loan prediction.